

Network Equilibrium of Coupled Transportation and Power Distribution Systems

Wei Wei, *Member, IEEE*, Lei Wu, *Senior Member, IEEE*, Jianhui Wang, *Senior Member, IEEE*, and Shengwei Mei, *Fellow, IEEE*

Abstract—This paper presents a holistic modeling framework for the interdependent transportation network and power distribution network. From a system-level perspective, on-road fast charging stations would simultaneously impact vehicle routing in the transportation system and load flows in the distribution system, therefore tightly couple the two systems. In this paper, a dedicated traffic user equilibrium model is proposed to describe the steady-state distribution of traffic flows comprised of gasoline vehicles and electric vehicles. It encapsulates route selections, charging opportunities, electricity prices, and individual rationalities of minimum travel expense in a convex traffic assignment problem over an extended transportation network. An adaptive path generation oracle is suggested to solve the problem in a tractable manner. Economic operation of the power distribution system is formulated as an alternating current optimal power flow problem. Convex relaxation is performed. The optimal generation dispatch and nodal electricity prices can be computed from a second-order cone program. It is revealed that an equilibrium state will emerge due to the rational behaviors in the coupled systems, which is characterized via a fixed-point problem. A best-response decomposition algorithm is suggested to identify the network equilibrium through iteratively solving the traffic assignment problem and the optimal power flow problem, both of which entail convex optimization. Illustrative examples are presented to validate related concepts and methods.

Index Terms—Electric vehicle, interdependency, locational marginal price (LMP), network equilibrium, optimal power flow, power distribution network, static traffic assignment, transportation network, Wardrop user equilibrium.

Manuscript received November 27, 2016; revised May 10, 2017; accepted June 26, 2017. Date of publication July 4, 2017; date of current version October 19, 2018. The work of W. Wei and S. Mei was supported by the National Natural Science Foundation of China under Grant 51621065. The work of L. Wu was supported by the U.S. National Science Foundation under Grant PFI-BIC-1534035. The work of J. Wang was supported by the U.S. Natural Science Foundation under Grant 1638348. Paper no. TSG-01669-2016. (*Corresponding author: Wei Wei.*)

W. Wei and S. Mei are with the State Key Laboratory of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China (e-mail: wei-wei04@mails.tsinghua.edu.cn; meisengwei@mail.tsinghua.edu.cn).

L. Wu is with the Department of Electrical and Computer Engineering, Clarkson University, Potsdam, NY 13699 USA (e-mail: lwu@clarkson.edu).

J. Wang is with the Department of Electrical Engineering, Southern Methodist University, Dallas, TX 75275 USA, and also with the Energy Systems Division, Argonne National Laboratory, Argonne, IL 60439 USA (e-mail: jianhui.wang@ieee.org).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TSG.2017.2723016

NOMENCLATURE

Major symbols and notations used throughout the paper are summarized below for quick reference. Others are defined following their first appearances in case of need.

Abbreviations

ACOPF	Alternating current optimal power flow.
BFM	Branch flow model.
BPR	Bureau of public roads.
DCOPF	Direct current optimal power flow.
EV	Electric vehicles.
FCS	Fast charging station.
GV	Gasoline vehicle.
KKT	Karush-Kuhn-Tucker.
LMP	Locational marginal price.
MILP	Mixed integer linear program.
NCP	Nonlinear complementarity program.
NLP	Nonlinear program.
NTE	Network equilibrium.
O-D	Origin-destination.
OPF	Optimal power flow.
PDN	Power distribution network.
SOC	State-of-charge.
SOCP	Second-order cone program.
TAP	Traffic assignment problem.
TN	Transportation network.
UE	User equilibrium.
V2G	Vehicle to grid.

Notations in the Traffic Assignment Problem

Sets

T_A^B	Set of bypass links.
T_A^C	Set of charging links.
T_A^R	Set of regular links.
T_A	Set of links in the TN, $T_A = T_A^R \cup T_A^C \cup T_A^B$.
T_N	Set of nodes in the TN.
D_T^{RS}	Set of O-D pairs.
k, r, s	This triplet denotes a path k between the origin $r \in T_N$ and destination $s \in T_N$, $(r, s) \in D_T^{RS}$.
K_g^{rs}	Index set of available paths for GVs between O-D pair $(r, s) \in D_T^{RS}$.
K_e^{rs}	Index set of available paths for EVs between O-D pair $(r, s) \in D_T^{RS}$.
$\mathbb{B}^n$	Set of n -dimensional vectors with binary elements.

$\mathbb{R}^{m \times n}$	Set of $m \times n$ matrices with real number entries.
$ Z $	The number of elements in set Z .
Parameters	
c_a	Capacity of link $a \in T_A$.
d	The charging availability vector.
E_B	Energy demand of each EV.
q_g^{rs}	Trip rate of GV between O-D pair (r, s) .
q_e^{rs}	Trip rate of EV between O-D pair (r, s) .
t_a^0	Free travel time on link $a \in T_A$.
Δ_G	The link-path incidence matrix for GV.
Δ_E	The link-path incidence matrix for EV.
Λ	The node-link incidence matrix of TN.
λ_a^j	LMP at charging link $a \in T_A^C$ connected to bus j .
ω	The monetary value of travel time.
Variables	
c_{kg}^{rs}	Travel expense of GV on path k between (r, s) .
c_{ke}^{rs}	Travel expense of EV on path k between (r, s) .
f_{kg}^{rs}	GV flow on path k between (r, s) .
f_{ke}^{rs}	EV flow on path k between (r, s) .
t_a	Vehicle travel time on link a .
u_g^{rs}	Minimal travel expense of GV between (r, s) .
u_e^{rs}	Minimal travel expense of EV between (r, s) .
x_a	Aggregated traffic flow on link a .
v_g	Binary vector used to identify active paths of GVs.
v_e	Binary vector used to identify active paths of EVs.

Notations in the OPF Problem

Sets and Indices

E_L	Set of lines in the PDN.
E_N	Set of buses in the PDN.
$\pi(j)$	Child buses of bus j .
l, i, j	This triplet denotes a distribution line $l \in E_L$ with its head bus $i \in E_N$ and tail bus $j \in E_N$.

Parameters

a_j, b_j	Energy production cost coefficients at bus j .
p_j^{dt}	Traditional active power demand at bus j .
p_j^{gn}, p_j^{gm}	Active power generation bounds at bus j .
q_j^d	Traditional reactive power demand at bus j .
q_j^{gn}, q_j^{gm}	Reactive power generation bounds at bus j .
r_{ij}^l	Resistance of line l connecting buses i and j .
S_l	Capacity of apparent power flow of line l .
U_j^n, U_j^m	Bounds of square voltage magnitude at bus j .
x_{ij}^l	Reactance of line l connecting buses i and j .
z_{ij}^l	Impedance of line l , $(z_{ij}^l)^2 = (r_{ij}^l)^2 + (x_{ij}^l)^2$.
ρ	Contract electricity price with the main grid.

Variables

I_{ij}^l	Square magnitude of current in line l .
p_j^{dc}	Active power demand of charging link $a \in T_A^C$ connected to bus $j \in E_N$.
p_j^d	Total active power demand at bus j .
p_j^g	Active power generation at bus j .
P_{ij}^l	Active power flow in line l .
q_j^g	Reactive power generation at bus j .
Q_{ij}^l	Reactive power flow in line l .
U_j	Square magnitude of voltage at bus j .

I. INTRODUCTION AND RELATED WORKS

THE EXCESSIVE consumption of fossil fuel has brought a suite of environmental, political, and economic problems. Cutting down the petrol usage has attracted wide attention in many countries. Electric vehicle (EV) is a promising alternative option for urban transportation due to its low environmental impact. The proliferation of EVs has been acknowledged as an effective remedy for mitigating environmental pollution and realizing sustainable development [1].

Despite their cleanliness, massive recharging of EVs in fast charging stations (FCSs) may create notable interaction between the transportation network (TN) and the power distribution network (PDN) [2]–[4]: Traffic congestion patterns and regulation policies in the TN will influence the driving patterns of EVs, which further impact spatial and temporal distributions of electrical loads as well as the operation of PDN; The queueing time and electricity prices in FCSs also affect travel plans of EVs and consequently vehicular flows in the TN. In this manner, traffic flows and electric power flows intertwine with each other through on-road charging facilities. Rational route choices of vehicles, the system-wide economic dispatch, and locational marginal prices (LMPs) of electricity precipitate such interdependency, which brings about both challenges and opportunities to the planning, operation, and control of the coupled infrastructures. In fact, both of TN and PDN have been the subject of decades of research. Related works are briefly summarized as follows.

A. Transportation System Research

The distinct feature of urban transportation compared with highway and railway transportation originates from congestions, which translate into time delays. The main target of urban transportation research has been focused on understanding and possibly alleviating congestions. In 1952, Wardrop pointed out the condition which leads to a stable traffic flow pattern in a congested TN [5], called a user equilibrium (UE), such that no traveler has the incentive to alter their routes unilaterally. Mathematical programming based approaches for computing the UE originated from 1956, when Beckmann showed that the Wardrop equilibrium condition is equivalent to the optimality condition of a strictly convex traffic assignment problem (TAP) [6]. A more thorough literature review on the early stage of UE studies can be found in [7]. Introduction to basic theory of traffic equilibria can be found in [8].

Early EV models usually suffer from limited battery capacities and driving ranges. They require more frequent recharging than refueling of gasoline vehicles (GVs), which inevitably restricts the route choices and abates the driving experiences.

To highlight the routing behavior of EVs, UE models with specific requirements of EVs are proposed. Most work in this line is found in the transportation research. For instance, a UE model with length constraints on EV paths is discussed in [9], and is further generalized in [10] to include trip chains. A basic mixed UE model with GVs and EVs is investigated in [11], and two efficient algorithms are suggested to solve the proposed model. A comprehensive UE model which jointly incorporates destination, route, and parking choices of GVs

and EVs subject to the driving range limit is presented in [12]. More dedicated UE models with battery EVs are proposed in [13]. Under different assumptions on the dependency of battery energy consumption and recharging time on traffic flows, three UE models are suggested in [13]; two of which are formulated as convex TAPs, and the third one yields a nonlinear complementarity problem (NCP).

In summary, the limited driving range of EV is the main concern of research community in this category. The impact from the power system side and the interdependency between the two networks are usually ignored.

B. Power System Research

Grid impacts of EV charging facilities have been the subject of power system research for a long time. Extensive literatures can be found, while some representative ones are introduced, which can be roughly categorized into three groups.

The first category is devoted to the profit-driven charging arbitrage. Bidirectional energy trading allows EVs to buy electricity from the grid when it is cheap and possibly sell it back when it becomes expensive. A decentralized coordinated charging method is developed in [14], in which EVs minimize their individual costs in response to real-time electricity prices, and the aggregator aims to maximize its profit and maintain system security. A two-stage stochastic program based risk-averse approach for self-scheduling of an EV aggregator is suggested in [15], which maximizes the aggregator's profit in day-ahead and balancing markets while considering uncertainties resulting from market prices and fleet mobility. A similar problem has been discussed in [16] using robust optimization, in which the aggregator's profit in the worst-case market price realization is maximized. The optimal charging strategy of plug-in electric taxis with maximum profit under uncertain electricity prices and time-varying incomes is investigated in [17], in which distinct features of EV taxis are considered. The optimal bidding strategy of a price-taker EV aggregator in the short-term energy and regulation markets is studied in [18]. A similar problem is considered in [19] by incorporating the market power of an aggregator, and is formulated as a bilevel optimization problem. Electricity price is determined from dual variables of the lower-level market clearing problem. A more comprehensive review on different charging schemes can be found in [20].

The second category aims to exploit V2G enabled ancillary services to support power system operations. For example, EV participation in the ramping market is considered in [21], and supplementary frequency regulation provision is discussed in [22]–[24]. EVs have the potential to facilitate renewable energy integration. They can offer storage capacities in multiple time scales to maintain a relatively smooth output level. In the aggregator-level research, a virtual battery model is proposed in [25] to simulate EV fleet charging and discharging behavior. Uncertainties of driving patterns and wind power are modeled by scenarios. A real-time energy management scheme for grid-connected charging parks is designed in [26] which helps shift peak demands and mitigate adverse impacts of EV charging on the power grid. EV parking lots are

modeled as storage devices in coupled heat-electricity systems with wind generation in [27]. The driving pattern uncertainty is translated into capacity and state-of-charge uncertainties in each hour. In the system-level research, coordinated scheduling of EV fleets, wind farms, and thermal units are studied in [28] and [29] via stochastic unit commitment, and in [30] via economic dispatch.

The third category endeavours to reveal grid impacts of aggregated EV charging from various perspectives. For instance, the effect of EV charging on the aging of distribution transformers is analyzed in [31]. A temperature-based smart charging method is suggested to alleviate possible damages. The impact of V2G on hourly generation schedules of conventional units is studied in [32], in which EV mobility is taken into account. Integration requirements, economic benefits, technical challenges, and charging strategies for V2G interfaces with their grid impacts are comprehensively reviewed in [33]. A stochastic model is proposed in [34] to simulate impacts of EV charging on system losses, voltage profile, and network congestions. The financial impacts of three charging strategies on distribution grids are compared in [35]. It is suggested that the minimum-peak charging method may be the best option from the system operation perspective.

In aforementioned research, a common assumption is that driving patterns of EVs are known, i.e., arrival rates and timings, as well as amounts of charging requests are specified either in a deterministic or a stochastic manner in advance, which can be determined from data-driven methods [36]–[38] or queueing theory [39], or modeled by a probabilistic model via Markov decision process [40]. This assumption is reasonable for aggregator-level studies, such as the management of a residential parking lot, or a single FCS. However, it may no longer be appropriate if the system-level interdependency is under investigation.

C. Interdisciplinary Research

To capture the system-level interaction, the network flows in TN and PDN should be jointly considered. The former is depicted by the UE model, whereas the latter usually gives rise to an optimal power flow (OPF) problem. Early attempt has been reported in the charging station planning issue. For example, by solving an OPF and a flow capturing problem coordinately, the service area of the FCS is enlarged, and the grid impact in term of net losses and voltage deviation is minimized [41]. The method is further improved in [42] and [43], in which the spatial distribution of vehicle flows in the TN is estimated from the UE, such that the rationality of individual travelers and the system-level interdependency is better portrayed. Due to its capability of modeling system interactions, this framework is utilized in [44] to study the operation of on-road wireless charging facilities in an electrified TN. A distributed algorithm is proposed based on alternating direction method of multipliers. Recently, there is an increasing number of articles on the charging facility and network expansion planning problem, interested readers are referred to [45]–[47],

TABLE I
COMPARISON OF MODELS IN RELEVANT LITERATURES

Ref.	Network model		Vehicle		Pricing model	
	TN	PDN	GV	EV	Road	Power
[48]	UE	DCOPF	×	—	×	LMP
[49]	UE	ACOPF	✓	—	Toll	LMP
[50]	UE	ACOPF	✓	—	Toll	×
[51]	SO	DCOPF	✓	✓	Toll	LMP
[44]	UE	DCOPF	✓	—	×	LMP

to name just a few, but their focus is generally different from the scope of this paper.

To further investigate the interplay between electricity prices and UE, an equilibrium model for the regional coupled TN and PDN infrastructures is proposed in [48], which integrates a stochastic UE problem, a direct current optimal power flow (DCOPF) problem, and a locational marginal pricing (LMP) scheme. The framework is then used to determine the sites of a given number of charging facilities. A similar paradigm is employed in [49] and [50] to study electricity price and road toll which will help improve the over-all operation condition of the coupled networks. A distributed pricing policy is suggested in [51], in which TN and PDN collaborate with each other in order to reach a social optimum operating point while keeping the data of each system private. In particular, the EV routing problem is considered as a shortest path problem on a modified TN, and the economic dispatch problem of the PDN is formulated as a DCOPF problem. A detailed comparison on the models in [44] and [48]–[51] has been summarized in Table I.

In summary, this category adopts non-atomic vehicular flow, which means that the system impact from the action of a single vehicle is infinitesimal. However, in [44] and [48]–[50], EVs are not adequately treated. A common assumption is that power demands in FCSs are proportional to road traffic flows. Although this assumption may be reasonable in certain cases, it is usually difficult to justify the charging rate coefficient, which has a decisive impact on the model of interdependency. An extended graphic model for the transportation system with both GV and EV is developed in [51] and incorporated in a system optimum problem, which is different from a UE problem where individual rationality is the driving force.

D. Contributions

This paper endeavours to build a comprehensive modeling framework that jointly considers vehicle routing, electricity pricing, and the system-level interaction brought by FCSs. It is revealed that the coupled network will reach an equilibrium state with three implications:

- 1) Individual rationality of single vehicles: In the TN, every vehicle travels on the route with minimal expense, yielding a UE pattern. Unilateral departure from the UE will increase the travel expense.
- 2) System rationality of the PDN operator: in the PDN, electricity is produced and distributed in the most

economical manner; consumers pay for their energy consumptions at the marginal cost.

- 3) Best response invariance: The UE pattern in the TN and LMP in the PDN constitute the best reaction of each other.

In these regards, a social optimum can be achieved. Main contributions of this paper are summarized as follows.

- 1) A multi-community UE model with the consideration of driving patterns, electricity prices, and FCSs. Different from UE models in [44] and [48]–[50], in this paper, EVs are treated separately like [51], which will give analytical expressions of electric loads in FCSs, and deeper insights on how FCS locations will influence EV routes. Unlike [48] and [49] which use a multi-nomial logit function to model destination choices (charging locations) of EVs, whose coefficients should be calibrated carefully, the origin-destination (O-D) pairs of EVs are fixed in this paper, and the battery is recharged en route, whose time consumption is accounted for in the travel expense, while routes and charging locations are simultaneously optimized according to the UE condition. Different from the UE models proposed in [9]–[13], range anxiety is not a major concern. The reason is explained in Section II-A. Instead, capacities of FCSs will be considered, which have been neglected in [48]–[50] and [9]–[13]. The proposed model may help alleviate long queues in charging stations. The UE is shown to be the optimal solution of a convex TAP problem over an extended TN, where a FCS is replaced by a special graphic module with nodes and links. In addition, we suggest a practical path generation oracle, which iteratively identifies a subset of paths that are most likely to be used. In this way, the TAP can be established without enumerating unnecessary paths, and solved in a more tractable manner with much fewer variables and constraints.
- 2) The network equilibrium (NTE) model comprised of a UE problem and an ACOPF problem based on the branch flow model (BFM) in [52]–[54]. From the transportation system side, UE characterizes the traffic flow distribution under individual rationality, and gives the electrical demands of FCSs; From the power system side, LMPs are determined from the system-level OPF solution, and passed to the UE problem, constituting a close-loop feedback, which is similar to [48]. The NTE boils down to a fixed-point problem. Its existence is guaranteed by Brouwer's fixed-point theorem under certain continuity assumptions. Stability and uniqueness are also discussed. The ACOPF based marginal pricing model is more accurate than the DCOPF based ones in [44], [48], and [51], because a branch in the PDN often has a large resistance to reactance ratio, and network losses are rigorously considered in the ACOPF model. In [49] and [50], there is a central agency which coordinates the operation of TN and PDN. In [51], a TN operator can guide routing decisions of vehicles towards a social optimum (SO). These assumptions are different from the market setting in this paper.

- 3) The best response decomposition algorithm based on convex optimization, which solves TAP and OPF iteratively till a fixed point is reached. Unlike [49] which solves ACOPF in its non-convex form using nonlinear programming (NLP) solvers, convex relaxation [55] is performed, and the ACOPF problem comes down to an SOCP. Meanwhile, the TAP has a strictly convex objective function over a polyhedral feasible region. Therefore, both of them can be readily solved. Convergence to a fixed point is guaranteed by the existence and stability conditions of NTE.

The remaining parts of this paper are organized as follows. The graphic model of the TN with FCSs, the multi-community UE problem and its corresponding convex TAP, as well as the delayed path generation algorithm are developed in Section II. A small example is given to illustrate related concepts for readers who are not familiar with transportation system models. ACOPF and LMP of the PDN is introduced in Section III. The fixed-point formulation, basic properties, and a best-response decomposition algorithm of the NTE are presented in Section IV. Case studies on a testing coupled network are provided in Section V. Conclusions are given in Section VI.

II. TRANSPORTATION NETWORK MODEL

A. Assumptions

Some observations, considerations, and basic assumptions are summarized to facilitate building the mathematical model.

- 1) All EVs in need of recharging service have the same battery size. When they leave from the origin, their batteries have the same initial state-of-charge (SOC). Although this assumption seems rather restrictive, it could be relaxed by incorporating several aggregated classes of EVs, and each class represents different battery status and charging requirements. A meticulous description on the heterogeneous battery information is difficult and also unnecessary in the system-level study. The single EV routing problem considering recharging plans has been thoroughly discussed in existing research, such as [56]–[58]. This assumption will be analyzed quantitatively in Item 3 below.
- 2) When traveling to their destinations, a GV seeks the route which has minimal travel time, and an EV looks for the route which minimizes its travel expense composed of the travel time and charging cost. EVs which do not need battery recharge are considered as GVs, as they have the same route selection criterion.
- 3) According to a recent survey [59], the maximum driving distances of current EV models vary from 85km to 586km. We envision that in the near future, the average maximum driving distance of EV can reach 300km, while a single trip in a city is typically less than 30km. Moreover, EV drivers may consider recharging their cars when the remaining driving range is 40km-50km. In this regard, we assume: a) An EV is able to reach any FCS without running out of energy. b) In a FCS, the battery will be fully recharged. c) According to the driving range data, the energy consumption before an EV arrives at

an FCS only accounts for a small fraction of the battery SOC, and thus can be neglected. Combined with Item 1 above, we further assume: d) All EVs receive the same amount of energy E_B (approximately 85% of the battery SOC) in an FCS. e) Chargers in one FCS have the same rated power, and the time for recharging a vehicle is proportional to E_B , therefore the same for all vehicles. In fact, similar to the discussion in Item 1, assumptions d)-e) can be relaxed by incorporating multiple EV groups categorized by different charging requests. Above settings make the recharging action of an EV similar to the refueling behavior of a GV, which is likely to happen in the near future. These assumptions will greatly simplify the model without significantly departure from the reality. Refueling of GVs can be modeled through the same way, but neglected in this paper because it does not directly impact the PDN.

- 4) The queueing time increases with the number of EVs visiting the FCS growing. The total time spent in the FCS is approximated by the Davidson function [60], which imposes an upper bound on the vehicular flow, reflecting the capacity of an FCS.
- 5) The distribution electricity market is cleared in accordance with an ACOPF model. Energy transaction is paid at the real-time LMP.

B. The Graphic Model

The foundation of transportation system modeling can be found in [8]. The basic model is extended to incorporate EVs and FCSs. A TN can be represented by a connected graph $G_T = [T_N, T_A]$, where T_N denotes the set of nodes, representing origins, destinations, and intersections; T_A denotes the set of links, representing lanes or roadway segments. The node-link incidence matrix $\Lambda \in \mathbb{M}^{|T_N| \times |T_A|}$ depicts network topology. Each column of Λ corresponds to a link and has two non-zero elements: 1 (−1) at the component associated with the entrance (exit) node.

Every vehicle in the TN leaves from an origin r and travels to its destination s . The GV (EV) flow q_g^{rs} (q_e^{rs}) between every O-D pair $(r, s) \in D_T^{RS}$ (also called the traffic demand) is known or can be predicted with relative high accuracy. Uncertain traffic demand is addressed in [61] via robust optimization. Each O-D pair (r, s) is connected by a set of paths, which are labeled by $k \in K_g^{rs}$ ($k \in K_e^{rs}$) for GVs (EVs). A path is a chain of connected links. The link-path incidence matrix $\Delta_G = [\Delta_G^{rs}], \forall (r, s)$ ($\Delta_E = [\Delta_E^{rs}], \forall (r, s)$) depicts the path topology, where $\Delta_G^{rs} \in \mathbb{M}^{|T_A| \times |K_g^{rs}|}$ ($\Delta_E^{rs} \in \mathbb{M}^{|T_A| \times |K_e^{rs}|}$) is the sub-matrix corresponding to O-D pair (r, s) . Its element $\delta_{akg}^{rs} = 1$ ($\delta_{ake}^{rs} = 1$) if path k of GV (EV) passes link a , and $\delta_{akg}^{rs} = 0$ ($\delta_{ake}^{rs} = 0$) otherwise.

FCSs are located beside lanes. In [48]–[50], they are not modeled separately. In order to simulate the choice of an EV to stop for getting charged or skip and keep driving, the FCS is represented by a two-node two-link module shown in Fig. 1. It consists of an entrance node I, an exit node E, a charging link $I \rightarrow \text{FCS} \rightarrow E$, and a bypass link $I \rightarrow E$ with zero travel time. Under this characterization, the TN with FCS can be

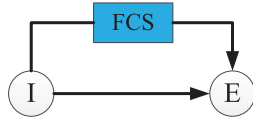


Fig. 1. Representation of a fast charging station.

consistently described as a traditional graph with only nodes and links. This description also provides a plausible way to evaluate the charging demand in a FCS, which is proportional to the EV flow in the charging link.

In an urban transportation system, congestion would dynamically change the fastest path between an O-D pair. The travel time on a particular link depends on the traffic flow it is carrying, which is a collective behavior resulting from rational route choices of all drivers. The congestion on a link is usually modeled by a latency function $t_a(x_a)$ [8]. In this paper, links are classified into three categories:

- 1) Regular link without a FCS (denoted by $a \in T_A^R$). The latency function $t_a(x_a)$ should be strictly increasing in x_a . The most widely used one is suggested by the Bureau of Public Roads (BPR) [62] and given by

$$t_a(x_a) = t_a^0 \left[1 + 0.15 \left(\frac{x_a}{c_a} \right)^4 \right], \quad \forall a \in T_A^R \quad (1)$$

where t_a^0 is the free travel time, i.e., the length of link a divided by the speed limit, and c_a is called the capacity. In fact, the capacity is not a mandatory limit for a regular link, instead, the travel time is penalized by a quick growth if $x_a > c_a$. Physically, c_a is the traffic flow on link a when $t_a = 1.15t_a^0$. BPR function (1) has been widely used in literature, say, [10]–[13], [42], [48]–[50].

- 2) Charging link with an FCS (denoted by $a \in T_A^C$), i.e., $I \rightarrow \text{FCS} \rightarrow E$ in Fig. 1. The time spent on link $a \in T_A^C$ consists of two parts: charging time and queueing time. The former is a constant t_a^0 which depends on the charging energy E_B and rated power of the charger, as discussed in Section II-A; the latter should be slightly increasing if $x_a < c_a$, and asymptotic to the capacity limit. To model such a situation, we employ the Davidson function which is developed in [60] based on the queueing theory, and expressed as

$$t_a(x_a) = t_a^0 \left[1 + J \left(\frac{x_a}{c_a - x_a} \right) \right], \quad \forall a \in T_A^C \quad (2)$$

where c_a represents the capacity of FCS on charging link $a \in T_A^C$, and parameter J controls the shape of the function. The smaller J is, the steeper $t_a(x_a)$ approaches to infinity when x_a tends to c_a .

- 3) Bypass link (denoted by $a \in T_A^B$), i.e., $I \rightarrow E$ in Fig. 1. As the length of the bypass link is very short, the travel time on a bypass link can be neglected, i.e.,

$$t_a(x_a) = 0, \quad \forall a \in T_A^B \quad (3)$$

In order to model different charging requests due to heterogeneous battery capacity and SOC, as well as multiple types of chargers with different levels of rated power, the charging

station module shown in Fig. 1 can be extended by employing multiple charging links. Each link serves an aggregated group of EVs that share the same values of E_B and t_a^0 .

C. Traffic Flow and Travel Expense Model

The concept of path plays a central role in the UE problem, which is defined below.

Definition 1: A feasible path labeled by $k \in K_g^{rs}$ ($k \in K_e^{rs}$) for GV (EV) between O-D pair (r, s) is a chain consisting of connected links from the origin r to the destination s , if it does not (does) visit one FCS. A path is called active (inactive or redundant) if it carries strictly positive (zero) traffic flow.

The feasible path depends on the network topology, and is independent of the congestion pattern and travel expenses. All feasible paths can be calculated and enumerated off-line. However, most of them are redundant and will not be visited at all. A special path generation oracle will be discussed later to identify active paths dynamically and obviate exhaustive path enumeration in large-scale applications. For now we consider that all feasible paths as well as the link-path incidence matrix Δ_G (Δ_E) are available, without jeopardizing the soundness of discussions.

Let f_{kg}^{rs} (f_{ke}^{rs}) represent the GV (EV) flow on path k between O-D pair (r, s) , with the help of Δ_G (Δ_E), the network flow constraints can be given as follows

$$x_a = \sum_{rs} \sum_{k \in K_g^{rs}} f_{kg}^{rs} \delta_{akg}^{rs} + \sum_{rs} \sum_{k \in K_e^{rs}} f_{ke}^{rs} \delta_{ake}^{rs}, \quad \forall a \in T_A \quad (4)$$

$$\left\{ \begin{array}{l} f_{kg}^{rs} \geq 0, \quad \forall k \in K_g^{rs}, \quad \sum_{k \in K_g^{rs}} f_{kg}^{rs} = q_g^{rs}, \quad \forall (r, s) \end{array} \right\} \quad (5)$$

$$\left\{ \begin{array}{l} f_{ke}^{rs} \geq 0, \quad \forall k \in K_e^{rs}, \quad \sum_{k \in K_e^{rs}} f_{ke}^{rs} = q_e^{rs}, \quad \forall (r, s) \end{array} \right\} \quad (6)$$

where equality (4) defines the relation between link flow and path flow; (5) and (6) are flow conservation conditions for GV and EV between each O-D pair.

Travel time t_{kg}^{rs} and travel expense c_{kg}^{rs} of GV on path k between O-D pair (r, s) are given as

$$t_{kg}^{rs} = \sum_{a \in T_A} t_a(x_a) \delta_{akg}^{rs}, \quad c_{kg}^{rs} = \omega t_{kg}^{rs}, \quad \forall k \in K_g^{rs}, \quad \forall (r, s) \quad (7)$$

where ω is the monetary value of travel time. For simplicity, we assume that road travelers share the same ω linking time and dollars. For GV, minimum travel expense is equivalent to minimum travel time.

Similarly, travel time t_{ke}^{rs} and travel expense c_{ke}^{rs} of EV on path k between O-D pair (r, s) can be expressed by

$$t_{ke}^{rs} = \sum_{a \in T_A} t_a(x_a) \delta_{ake}^{rs}, \quad \forall k \in K_e^{rs}, \quad \forall (r, s) \quad (8)$$

$$c_{ke}^{rs} = \omega t_{ke}^{rs} + \lambda_a^j E_B, \quad \forall k \in K_e^{rs}, \quad \forall (r, s) \quad (9)$$

where λ_a^j is the LMP at charging link $a \in T_A^C$ connected to bus $j \in E_N$ in the PDN (a feasible path of EV visits only one FCS according to Definition 1), E_B is the received energy, which is a constant according to discussions in Section II-A.

The ACOF model and definition of LMP will be discussed in Section III.

D. The Mixed UE Problem

Network flow constraints (4)-(6) give all feasible traffic flow patterns in the TN. It is apparent that if anyone can cut down his travel expense by using another path, the flow pattern will change. Because of the rationality of minimum travel expense, a stable state emerges when travel expenses on all active paths are equal. The stable traffic flow pattern is also called a UE. The condition for UE is formally given below.

Proposition 1: The traffic flow reaches a UE if travel expenses of GVs (EVs) on all active paths between any given O-D pair are equal, and no greater than those which would be incurred on any unused path.

Proposition 1 is a generalization of Wardrop UE principle in [8] in the presence of EVs and FCSs. Each O-D pair is associated with two minimal travel expenses: u_g^{rs} for GVs, and u_e^{rs} for EVs. With this notation, Proposition 1 can be expressed in a logic form:

$$\begin{aligned} \text{if } f_{kg}^{rs} > 0, \text{ then } c_{kg}^{rs} &= u_g^{rs}, \forall k \in K_g^{rs}, \forall (r, s) \\ \text{if } f_{kg}^{rs} = 0, \text{ then } c_{kg}^{rs} &\geq u_g^{rs}, \forall k \in K_g^{rs}, \forall (r, s) \end{aligned} \quad (10)$$

$$\begin{aligned} \text{if } f_{ke}^{rs} > 0, \text{ then } c_{ke}^{rs} &= u_e^{rs}, \forall k \in K_e^{rs}, \forall (r, s) \\ \text{if } f_{ke}^{rs} = 0, \text{ then } c_{ke}^{rs} &\geq u_e^{rs}, \forall k \in K_e^{rs}, \forall (r, s) \end{aligned} \quad (11)$$

which can be formulated as the following NCP

$$0 \leq f_{kg}^{rs} \perp c_{kg}^{rs} - u_g^{rs} \geq 0, \forall k \in K_g^{rs}, \forall (r, s) \quad (12)$$

$$0 \leq f_{ke}^{rs} \perp c_{ke}^{rs} - u_e^{rs} \geq 0, \forall k \in K_e^{rs}, \forall (r, s) \quad (13)$$

$$\text{Cons-Flow} = \{ (4), (5), (6) \} \quad (14)$$

$$\text{Cons-Cost} = \{ (7), (8), (9) \} \quad (15)$$

where notation $0 \leq a \perp b \geq 0$ stands for complementarity and slackness conditions $a \geq 0, b \geq 0$, and $ab = 0$. That is, at most one of a and b can be strictly positive. However, it is generally difficult to compute the UE by directly solving NCP (12)-(15) via a general purpose NLP solver, because standard constraint qualifications are violated due to the complementarity and slackness conditions in (12) and (13). A thorough discussion can be found in [63]. The MILP reformulation in [50] can be applied if the latency functions $t_a(x_a)$ in c_{kg}^{rs} and c_{ke}^{rs} are represented by piece-wise linear functions. However, it is not preferred because the UE routine will be called repeatedly while computing the NTE, and solving large-scale MILPs may be time-consuming.

Following the paradigm in [6] and [8], it turns out that NCP (12)-(15) constitute the KKT conditions of the following optimization problem, which is called a TAP

$$\min\{F_{TAP} \mid \text{s.t. Cons-Flow}\} \quad (16)$$

where the objective function F_{TAP} is

$$\sum_{a \in T_A^R} \int_0^{x_a} \omega t_a(\theta) d\theta + \sum_{a \in T_A^C} \left(\int_0^{x_a} \omega t_a(\theta) d\theta + \lambda_a^j E_B x_a \right)$$

For the BPR function, the integration can be calculated as

$$\int_0^{x_a} t_a(\theta) d\theta = t_a^0 \left(x_a + \frac{0.03x_a^5}{c_a^A} \right), \forall a \in T_A^R \quad (17)$$

and for the Davidson function, we have

$$\begin{aligned} \int_0^{x_a} t_a(\theta) d\theta &= t_a^0 (1 - J) x_a \quad \forall a \in T_A^C \\ &+ t_a^0 c_a J \ln \left(\frac{c_a}{c_a - x_a} \right), \quad x_a < c_a \end{aligned} \quad (18)$$

In either case, $t_a(x_a)$ is differentiable and strictly increasing, hence it is clear that the objective function F_{TAP} is convex, because

$$\frac{d^2}{dx_a^2} \int_0^{x_a} t_a(\theta) d\theta = \frac{dt_a(x_a)}{dx_a} > 0, \quad \forall a \in T_A^R \cup T_A^C$$

Convex TAP (16) with linear constraints can be globally solved by traditional NLP solvers. Its optimal solution x_a^* , $\forall a$ determines the link flow pattern at the UE.

Assumption 1: The TN is strongly connected: there is at least one feasible path for an EV to reach each FCS regardless of its origin and destination, and the total charging capacity is adequate, i.e., $\sum_{rs} q_e^{rs} < \sum_{a \in T_A^C} c_a$.

Assumption 1 ensures that TAP (16) is always feasible. Real TNs usually satisfy this assumption, so at least one UE exists for such systems.

Uniqueness is guaranteed by the strict convexity of F_{TAP} in link flow variables. It should be mentioned that although F_{TAP} is not strictly convex in bypass link flows x_a , $\forall a \in T_A^B$, as they do not appear in the objective function, these variables can be eliminated by transformation $x_a = x_{r(a)} - x_{c(a)}$, where $x_{r(a)}$ is the total traffic flow on the regular link connected to the entrance node of the bypass link a , and $x_{c(a)}$ is the EV flow on the charging link corresponding to the bypass link a . In this regard, TAP (16) is equivalent to a strictly convex optimization problem with linear constraints, whose solution is unique.

E. A Practical Algorithm

Despite the convexity of TAP (16), enumerating all possible paths could be exhaustive for a large-scale TN. This issue has been discussed in [13]. The central idea is to identify a subset of paths which are most likely to be active. In this paper, we achieve this target via optimization. The development begins with the mathematical description of Definition 1.

Proposition 2: All possible paths for GVs and EVs between O-D pair (r, s) can be expressed by

$$V_g^{rs} = \left\{ v_g \mid \Lambda v_g = I^{rs}, d^T v_g = 0, v_g \in \mathbb{B}^{|T_A|} \right\} \quad (19)$$

$$V_e^{rs} = \left\{ v_e \mid \Lambda v_e = I^{rs}, d^T v_e = 1, v_e \in \mathbb{B}^{|T_A|} \right\} \quad (20)$$

where $I^{rs} \in \mathbb{R}^{|T_N|}$ has two non-zero elements, 1 and -1 at the entries corresponding to the origin node r and the destination node s ; $d \in \mathbb{B}^{|T_A|}$ is the charging availability vector, whose element is 1 for $a \in T_A^C$, and 0 otherwise. In view of the definition of Λ , $v_g \in V_g^{rs}$ ($v_e \in V_e^{rs}$) represents a chain of connected links from r to s which does not (does) visit one FCS. Let $\Delta_G^{rs} = [v_g^i]$, $\forall v_g^i \in V_G^{rs}$ and $\Delta_E^{rs} = [v_e^i]$, $\forall v_e^i \in V_E^{rs}$, then the link-path incidence matrices can be expressed as $\Delta_G = [\Delta_G^{rs}]$, $\forall (r, s) \in D_T^{RS}$ and $\Delta_E = [\Delta_E^{rs}]$, $\forall (r, s) \in D_T^{RS}$.

Algorithm 1 : Adaptive Path Generation

- 1: Let $x_a = 0, \forall a$, compute p_g^a and p_e^a , solve problems (22) and (23) for each O-D pair, and then build initial link-path incidence matrices Δ_G and Δ_E which will make TAP (16) feasible.
- 2: Solve the restricted TAP (16) with current Δ_G and Δ_E , the UE is $x_a^*, \forall a$.
- 3: With the obtained UE, update travel expenses p_g^a and p_e^a on each link; find minimal travel expenses u_g^{rs} and u_e^{rs} between (r, s) , and solve (22) and (23) for each O-D pair, the optimal solutions are v_g^{rs} and v_e^{rs} , and the optimal values are u_{gc}^{rs} and u_{ec}^{rs} .
- 4: If $u_{gc}^{rs} \geq u_g^{rs}, u_{ec}^{rs} \geq u_e^{rs}, \forall (r, s)$, terminate and report the current UE; otherwise, if $u_{gc}^{rs} < u_g^{rs}$ for some (r, s) , $\Delta_G^{rs} \leftarrow [\Delta_G^{rs}, v_g^{rs}]$; if $u_{ec}^{rs} < u_e^{rs}$ for some (r, s) , $\Delta_E^{rs} \leftarrow [\Delta_E^{rs}, v_e^{rs}]$; Update Δ_G and Δ_E , and go to step 2.

To obviate path enumeration, we first solve TAP (16) with a subset of available paths in K_g^{rs} and K_e^{rs} which will give a feasible TAP, then check whether the travel expense can be further reduced. To this end, suppose the current UE is $x_a^*, \forall a$, and corresponding link travel expense can be calculated as

$$p_g^a = \omega t_a(x_a^*), p_e^a = \omega t_a(x_a^*) + f_e^a, \forall a \in T_A \quad (21)$$

where the potential charging cost is $f_e^a = \lambda_a^j E_B, \forall a \in T_A^C$, and $f_e^a = 0, \forall a \in T_A^R \cup T_A^B$. The minimal travel expenses u_g^{rs} and u_e^{rs} between O-D pair (r, s) can be readily evaluated through (10) and (11) given the traffic flow $x_a^*, \forall a$.

To find a possibly better path between (r, s) for GVs and EVs, we solve the following two MILPs

$$u_{gc}^{rs} = \min_{v_g \in \mathbb{B}^{|T_A|}} \left\{ \sum_a p_g^a v_g^a \mid \Lambda v_g = I^{rs}, d^T v_g = 0 \right\} \quad (22)$$

$$u_{ec}^{rs} = \min_{v_e \in \mathbb{B}^{|T_A|}} \left\{ \sum_a p_e^a v_e^a \mid \Lambda v_e = I^{rs}, d^T v_e = 1 \right\} \quad (23)$$

Objective functions of (22) and (23) represent travel expenses perceived by a dummy vehicle (thus the traffic flow remains the same) on a feasible path restricted in V_g^{rs} and V_e^{rs} . If $u_{gc}^{rs} < u_g^{rs}$ ($u_{ec}^{rs} < u_e^{rs}$), the solution v_g (v_e) indicates a better path for GVs (EVs). A procedure for solving TAP (16) without path enumeration is shown in Algorithm 1. Since (22) and (23) identify favorite paths of GVs and EVs, Algorithm 1 always converges in a few iterations with affordable computational resources.

F. An Illustrative Example

A small example is given in this section to illustrate related concepts. The topology of the TN is shown in Fig. 2, which includes 2 nodes (T1, T2), 3 links, 1 O-D pair, and 2 FCSs (C1, C2). The extended network topology is given in the same figure, which is comprised of 6 nodes and 9 links, including 5 regular links, 2 charging links, and 2 bypass links. Link data are provided beside each branch. The base value of traffic flow is 100 vehicles/hour. The total traffic demand is 100p.u. EV accounts for 10% of the total traffic demand, and half of them

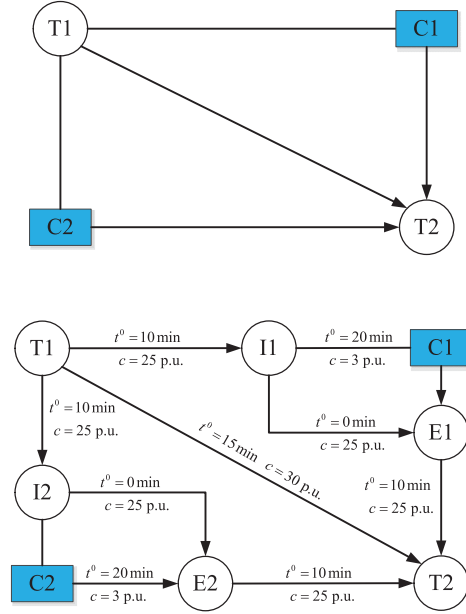


Fig. 2. Topology of a simple TN with FCSs.

need recharging. This setting leads to $q_g^{rs} = 95$ and $q_e^{rs} = 5$. The energy demand of each EV is $E_B = 50\text{kWh}$. The fixed charging time is 20 minutes. In Davidson function (2), $J = 0.05$ is adopted.

The node-link incidence matrix can be written as

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & -1 \end{bmatrix} \begin{matrix} \text{T1} \\ \text{I1} \\ \text{E1} \\ \text{I2} \\ \text{E2} \\ \text{T2} \end{matrix}$$

where each row (column) of Λ corresponds to a particular node (link). For example, the last row corresponds to link T1-T2. All available paths in this simple network can be enumerated. The link-path incidence matrices Δ_G and Δ_E are given by

$$\Delta_G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \Delta_E = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{matrix} \text{T1-I1} \\ \text{I1-C1-E1} \\ \text{I1-E1} \\ \text{E1-T2} \\ \text{T1-I2} \\ \text{I2-C2-E2} \\ \text{I2-E2} \\ \text{E2-T2} \\ \text{T1-T2} \end{matrix}$$

where each row (column) of Δ_G (Δ_E) is associated with a particular link (path). For example, the first column of Δ_E stands for the path T1→I1→C1→E1→T2. Please be aware that C1 is not a node, and C1 is emphasized for distinguishing this path from T1→I1→E1→T2 which is feasible for GVs and depicted by the first column of Δ_G .

We consider the UE in two scenarios. In the first scenario, we assume the electricity price is $\lambda = 100\$/\text{MWh}$ in both FCSs. TAP (16) is solved by Baron [64], and the UE is

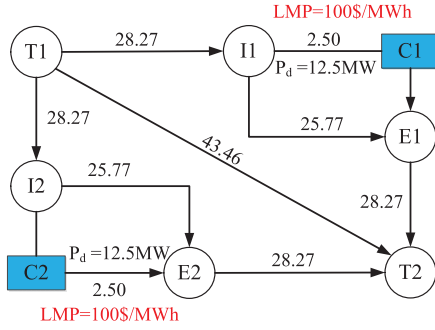


Fig. 3. Stable traffic flow pattern in the first scenario.

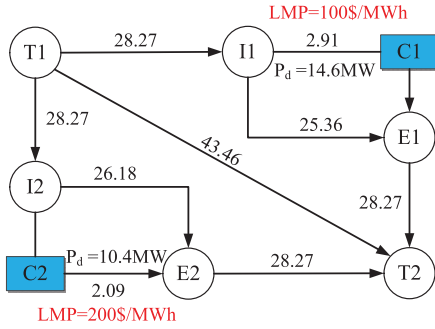


Fig. 4. Stable traffic flow pattern in the second scenario.

TABLE II
PATH TRAVEL TIMES AND EXPENSES

		Scenario 1		Scenario 2	
		Time (min)	Expense (\$)	Time (min)	Expense (\$)
GV-Path	T1-I1-E1-T2	24.91	4.1511	24.91	4.1511
	T1-I2-E2-T2	24.91	4.1511	24.91	4.1511
	T1-T2	24.91	4.1511	24.91	4.1511
EV-Path	T1-I1-C1-E1-T2	49.91	13.318	77.20	17.867
	T1-I2-C2-E2-T2	49.91	13.318	47.20	17.867

exhibited in Fig. 3. It is clear that the traffic flow is symmetrically distributed since the network itself (including electricity price) is symmetric. The aggregated charging power in FCSs can be calculated as $p_1^{dc} = x_a E_B = 12.5 \text{ MW} = p_2^{dc}$. Travel times and expenses on available paths are shown in Table II. We can see that travel expenses of GV (EVs) on each path between O-D pair (T1, T2) are equal, which is consistent with Proposition 1. Although the expense of EV is higher than that of GV due to the waiting time and cost for recharging, we do not include the refueling cost of GV. With the progress in advanced battery technology, the driving range of EV may catch up with that of GV in the near future. The average cost of EV should be lower owing to different prices of electricity and petrol.

To show the impact of electricity price on the route choice of EVs, the price in C2 is doubled. The new UE is portrayed in Fig. 4, from which we can see that more EV drivers prefer to charge their cars in C1 where the electricity is cheaper. Power demands in C1 and C2 are $p_1^{dc} = 14.6 \text{ MW}$ and $p_2^{dc} = 10.4 \text{ MW}$, respectively. Moreover, GV responds to the changes in EV flow pattern, and the total traffic flow is still symmetric. Certainly, prices of electricity will be affected

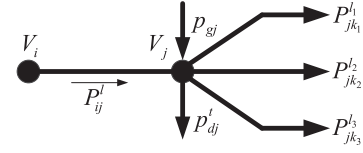


Fig. 5. Typical connection of a radial PDN.

by power demands of FCSs, which is neglected in this case but will be considered in the following sections.

It is interesting to notice that link traffic flows except those in the FCS modules are consistent in both scenarios. The phenomenon is easy to understand: at a UE pattern, the travel times on all GV feasible paths are equal. When a small perturbation is imposed on the electricity price, the small change of EV flow in FCS $a \in T_A^C$ will be compensated by the change of the traffic flow in the bypass link associated with that FCS, such that the external flow remains a UE pattern. However, this is clearly not the case in an EV dominant TN because there is not enough GVs to compensate the change in the EV flow.

III. DISTRIBUTION NETWORK MODEL

FCSs are connected to electric buses of a PDN. The PDN can be depicted by a graph $G_E = [E_N, E_L]$, where E_N and E_L denote sets of buses and distribution lines, respectively. The typical connection of branches in a radial PDN is shown in Fig. 5. Radiality has two implications: 1) each bus (except the slack bus) has a unique parent bus; 2) numbers of buses (excluding the slack bus) and lines are equal. Without loss of generality, we assume each bus connects to one generator and serves one traditional load with a fixed demand. Line flows and bus voltages in the PDN are amenable to BFM proposed in [52]–[54]

$$P_{ij}^l + p_j^g - r_{ij}^l I_{ij}^l = \sum_{k \in \pi(j)} P_{jk}^l + p_j^d, \quad \forall l \in E_L \quad (24)$$

$$Q_{ij}^l + q_j^g - x_{ij}^l I_{ij}^l = \sum_{k \in \pi(j)} Q_{jk}^l + q_j^d, \quad \forall l \in E_L \quad (25)$$

$$U_j = U_i - 2(r_{ij}^l P_{ij}^l + x_{ij}^l Q_{ij}^l) + (z_{ij}^l)^2 I_{ij}^l, \quad \forall l \in E_L \quad (26)$$

$$I_{ij}^l U_i = (P_{ij}^l)^2 + (Q_{ij}^l)^2, \quad \forall l \in E_L \quad (27)$$

where (24) and (25) are nodal active and reactive power balancing conditions; (26) is the forward voltage drop equation; (27) defines the apparent power at the head node of each line. The total active power demand is given by

$$p_j^d = p_j^{dt} + p_j^{dc}, \quad p_j^{dc} = x_a^j E_B, \quad \forall j \in E_N \quad (28)$$

where x_a^j is the EV flow passing charging link $a \in T_A^C$, and the corresponding FCS is connected to bus j in the PDN; E_B is the energy demand of each EV. BFM is shown to be equivalent to the classical bus-injection based power flow model for radial networks [65], but is more concise for PDNs, because all variables are directly measurable, and security limits can boil down to simple boundary constraints as follows

$$p_j^{gn} \leq p_j^g \leq p_j^{gm}, \quad \forall j \in E_N \quad (29)$$

$$q_j^{gn} \leq q_j^g \leq q_j^{gm}, \quad \forall j \in E_N \quad (30)$$

$$U_j^n \leq U_j \leq U_j^m, \forall j \in E_N \quad (31)$$

$$\sqrt{(P_{ij}^l)^2 + (Q_{ij}^l)^2} \leq S_l, \forall l \in E_L \quad (32)$$

$$P_{ij}^l - r_{ij}^l I_{ij}^l \geq 0, \forall l \in E_L \quad (33)$$

$$Q_{ij}^l - x_{ij}^l I_{ij}^l \geq 0, \forall l \in E_L \quad (34)$$

where (29)-(31) represent active and reactive generation capacities, as well as square voltage magnitude bounds at bus j ; (32) limits the apparent power flow in each line; (33) and (34) prevent reverse power flows on both sides of a distribution line, due to relay protection requirements of the PDN. For active distribution networks which allow reverse power flows, these two constraints can be relaxed.

The ACOPF problem of the PDN can be expressed as

$$\min \sum_{j \in E_N} \left[a_j (p_j^g)^2 + b_j p_j^g \right] + \rho \sum_{j \in \pi(0)} P_{0j}^l \quad (35)$$

$$\text{s.t. Cons-BPF, Cons-BND} \quad (36)$$

where Cons-BPF and Cons-BND include (24)-(28) and (29)-(34), respectively. The first term in objective function (35) is the production cost of local units in the PDN. P_{0j}^l denotes the active power delivered through line l connected to the slack bus, and ρ is the contract energy price with the main grid, hence the second term is the payment for purchasing electricity.

In (36), all constraints define convex regions except (27). By performing the convex relaxation method in [55], quadratic equality (27) is replaced with inequality $I_{ij}^l U_i \geq (P_{ij}^l)^2 + (Q_{ij}^l)^2$, whose canonical second-order cone form is

$$\left\| \begin{bmatrix} 2P_{ij}^l \\ 2Q_{ij}^l \end{bmatrix} \right\| \leq I_{ij}^l + U_i, \forall l \in E_L \quad (37)$$

and the convex ACOPF is denoted as

$$\min \sum_{j \in E_N} \left[a_j (p_j^g)^2 + b_j p_j^g \right] + \rho \sum_{j \in \pi(0)} P_{0j}^l \quad (38)$$

$$\text{s.t. Cons-CBPF, Cons-BND}$$

where Cons-CBPF is obtained from Cons-BPF by replacing (27) with (37).

We demonstrate that the convex relaxation performed on (27) is exact: at the optimal solution of problem (38), equality holds for every constraint in (37). To see this, suppose the power $p_0^g = \sum_{j \in \pi(0)} P_{0j}^l$ is supplied by a dummy generator connected to the slack bus, and its cost function is a linear function ρp_0^g , then ACOPF (38) and the OPF problem considered in [66] share the same form. As reverse flow is prohibited by (33)-(34), according to [66, Th. 2], the convex relaxation is exact if the original OPF problem (35)-(36) is feasible. A more thorough discussion on system parameters that will lead to an exact convex relaxation model is provided in [67]. For active distribution networks in which reverse power flow may exist, it is revealed in [68] that the SOCP relaxation will be still exact if active and reactive power flows do not become reverse simultaneously. Even if the relaxation is inexact, a sequential convex optimization approach proposed in [69] could recover an OPF

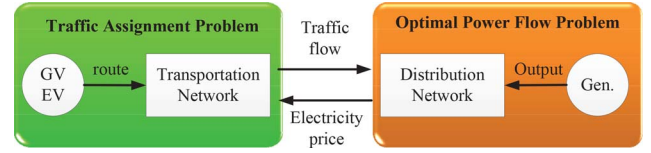


Fig. 6. Interdependency between the coupled networks.

solution that satisfies (36). With this exactness guarantee, we can retrieve LMPs through dual variables.

Definition 2: LMP λ_a^j (real-time electricity price) at charging link $a \in T_A^C$ connected to bus $j \in E_N$ is the dual variable associated with the active power balancing equation (24).

There are three ways to retrieve LMP:

- 1) Some commercial solvers offer primal and dual optimal solutions at the same time, such as Mosek [70].
- 2) Solve the dual problem of the relaxed OPF problem (38), which is also an SOCP, and extract related dual variables.
- 3) Substituting the optimal solution of the relaxed OPF problem (38) into the KKT optimality condition of the original OPF problem (35)-(36), giving rise to a set of linear equations in dual variables. LMP can be easily calculated by solving the set of linear equations.

IV. EQUILIBRIUM OF THE COUPLED NETWORK

A. Mathematical Formulation

The interdependency between TN and PDN is shown in Fig. 6: at the TN side, EVs plan their routes according to road congestion levels and electricity prices in FCSs, the vehicular flow passing each FCS gives the electrical demand; at the PDN side, given system demands including those of FCSs, LMPs are determined from ACOPF which in turn influence the traffic flow distribution. Under this interplay, an equilibrium state will emerge in the coupled networks. The equilibrium encapsulates the UE pattern in TN where no vehicle is able to reduce its travel expense by changing the route unilaterally, as well as the OPF state in PDN such that electricity is produced and delivered in the most economic manner, while consumers pay for their energy consumptions at marginal costs.

Let $\mathbf{x}^T$ be the vector of variables in TAP (16), and $\mathbf{x}^E$ be the vector of primal and dual variables in OPF problem (35)-(36), vector $\mathbf{x} = [\mathbf{x}^T; \mathbf{x}^E]$. The mapping $\mathcal{S}_{TAP}(\mathbf{x})$ stands for the UE solution under given LMPs, and $\mathcal{S}_{OPF}(\mathbf{x})$ represents the OPF solution and corresponding LMP under the given UE. Denote mapping $\mathcal{S}_{NTE}(\mathbf{x}) = [\mathcal{S}_{TAP}(\mathbf{x}); \mathcal{S}_{OPF}(\mathbf{x})]$, then we can define the network equilibrium.

Definition 3: The network equilibrium of the coupled TN and PDN is the solution of the following fixed-point problem

$$\mathbf{x} = \mathcal{S}_{NTE}(\mathbf{x}) \quad (39)$$

One possible way to express the NTE in a mathematical programming formulation is the KKT optimality condition based approach, which has a close relationship with the variational inequality theory. Let $\text{OPF-KKT}(\mathbf{x})$ be the KKT condition of OPF problem (35)-(36), in which $\mathbf{x}^T$ (charging power p_j^{dc}

appeared in (28)) is the parameter and $\mathbf{x}^E$ is the decision variable; let TAP-KKT($\mathbf{x}$) be the UE condition (12)-(15), in which $\mathbf{x}^E$ (LMP λ_a^j appeared in the travel expense (9)) is the parameter and $\mathbf{x}^T$ is the decision variable. Then, the NTE boils down to the stationary point of an NCP

$$\begin{cases} \text{TAP-KKT}(\mathbf{x}) \\ \text{OPF-KKT}(\mathbf{x}) \end{cases} \quad (40)$$

Due to the presence of numerous complementarity and slackness conditions, solving NCP (40) would be extremely challenging, if not impossible.

Before proceeding to the algorithm, we reveal some basic properties of the NTE. To facilitate understanding, some notations are clarified here. In the TAP, $x^L = [x_a], \forall a$ denotes the link flow vector; $f_G^P = [f_{kg}^{rs}], \forall (r, s), \forall k \in K_G^{rs}$ and $f_E^P = [f_{ke}^{rs}], \forall (r, s), \forall k \in K_E^{rs}$ are path flow vectors for GVs and EVs, respectively, vector $f^P = [f_G^P; f_E^P]$.

The feasible region of TAP is defined as

$$\mathcal{F}^{TAP} = \{ \mathbf{x}^T = (x^L, f^P) \mid \text{Cons-Flow} \}$$

Since all constraints are linear and all variables are bounded, $\mathcal{F}^{TAP}$ is a bounded polytope. The feasible region of link flow is the projection of $\mathcal{F}^{TAP}$ on the x^L -space:

$$\mathcal{F}^{TL} = \{ x^L \mid \exists f^P : (x^L, f^P) \in \mathcal{F}^{TAP} \} \quad (41)$$

$\mathcal{F}^{TL}$ is also a bounded polytope [71] (see Appendix B.2), thus convex and compact.

Define the FCS load injection vector $p^{dc} = [p_j^{dc}], \forall j$, where

$$p_j^{dc} = \begin{cases} x_a^j E_B, & \text{if bus } j \text{ serves FCS } a \in T_A^C \\ 0, & \text{if no FCS connects to bus } j \end{cases} \quad (42)$$

and load injection space P^{DC} from FCSs is

$$P^{DC} = \{ p^{dc}(x^L) \mid \forall x^L \in \mathcal{F}^{TL} \} \quad (43)$$

where $p^{dc}(x^L)$ stands for the relation between p^{dc} and x^L shown in (42).

1) *Existence*: We demonstrate the existence of an NTE by using Brouwer fixed-point theorem. To construct a continuous self-mapping, we make further assumptions on the UE and the OPF problem.

Assumption 2: The UE is continuous with respect to the electricity price at each FCS.

This assumption generally holds true if P^{DC} is not very large. To see this, a UE exists if Assumption 1 is met, which is independent of the electricity price. Moreover, From Proposition 1 we can image that path flows should be continuous in the electricity price, because all travel expense functions in (9) are continuous over $\mathcal{F}^{TL}$ and linearly depend on the LMP λ_a^j , hence a small change in the charging cost would lead to a small change in the path flow f^P . Finally, according to (4), the link flow x^L is a linear function of path flow f^P , and thus also continuous in the electricity price.

Assumption 3: OPF problem (35)-(36) is feasible for any given $p^{dc} \in P^{DC}$; electricity prices at FCS buses are continuous with respect to p^{dc} on P^{DC} .

The feasibility requirement can be satisfied as long as generation capacities, line flow limits, and bus voltage intervals are properly assigned. However, the continuity of LMP is generally not guaranteed. For example, the DCOPF based LMP is stepwise constant [72]. In this paper, since the convex relaxation is exact, LMP can be viewed as the optimal solution of the dual SOCP of problem (38). Local continuity of optimal solution function is studied in [73]. We can expect that LMP should be piecewise continuous. Nevertheless, continuity can be expected in the following two situations:

- 1) The set P^{DC} is small enough. This is true when p_j^{dc} only accounts for a small portion of the total demand.
- 2) A continuous pricing policy is enforced, such as the continuous LMP scheme described in [72].

Proposition 3: At least one network equilibrium exists if Assumptions 2-3 are satisfied.

Assumption 2 implies that $\mathcal{S}_{TAP}(\mathbf{x})$ is a continuous mapping. The mapping $x \mapsto p^{dc}$ from link flow to load injection shown in (28) and (43) is linear, thus continuous. Combined with Assumption 3, $\mathcal{S}_{OPF}(\mathbf{x})$ is a continuous mapping. Consider the following mapping: give a link flow pattern $x^L \in \mathcal{F}^{TL}$, first solve relaxed OPF problem (38) and retrieve the LMP, then solve TAP (16) with the obtained LMP, and the outcome is again a link flow pattern $x^L \in \mathcal{F}^{TL}$, which can be expressed as a composite self-mapping

$$\mathcal{M} : \mathcal{F}^{TL} \rightarrow \mathcal{F}^{TL} = \mathcal{S}_{TAP}(\mathcal{S}_{OPF}(\mathbf{x}))$$

which is continuous from the above analysis. Since polytope $\mathcal{F}^{TL}$ is convex and compact, according to Brouwer's fixed point theorem ([74, Corollary 6.6]), $\mathcal{S}_{TAP}(\mathcal{S}_{OPF}(\cdot))$ has a fixed point in $\mathcal{F}^{TL}$, which is the traffic flow pattern at the NTE. OPF variables at the NTE can be calculated from $\mathcal{S}_{OPF}(\mathbf{x})$. Actually, continuity is merely a sufficient condition to guarantee the existence of NTE. Intuitively, an NTE exists as long as nodal price and demand curves have an intersection, but are not necessarily continuous.

2) *Stability*: The NTE is locally stable, if

$$\frac{\partial \lambda_a^j}{\partial p_j^{dc}} > 0, \quad \frac{\partial \lambda_a^j}{\partial p_j^{dc}} \left| \frac{\partial x_a}{\partial \lambda_a^j} \right| E_B < 1, \quad \forall a \in T_A^C \quad (44)$$

holds at the equilibrium.

The stability of the NTE can be understood in the following way. First, a UE in the TN is stable according to its definition. Second, condition (44) indicates that when more EVs receive charging services on link $a \in T_A^C$, the electricity price λ_a^j at that FCS rises as p_j^{dc} grows, so does the travel expense of EV. Consequently, some EVs will turn to use other FCSs in light of Proposition 1, which would result in declines of x_a and p_j^{dc} . The second inequality in (44) ensures that $\mathcal{S}_{TAP}(\mathcal{S}_{OPF}(\cdot))$ is locally contractive. This mechanism maintains stability of the NTE. More importantly, under the rationality of minimum travel expense, EV drivers attempt to circumvent expensive charging services, which prevents demands and LMPs in those FCSs from rising higher. This adaptivity of UE creates system-wide demand responses which are helpful to PDN operation.

Algorithm 2 : Best Response Decomposition

- 1: Choose a convergence tolerance $\varepsilon > 0$, an allowed number of iterations K_{it}^M , and an initial LMP vector $\lambda \in \mathbb{R}^{|E_N|}$. The current iteration number is $k = 1$. Let $x^L = \mathbf{0}^{|T_A|}$.
- 2: Solve TAP (16) with fixed electricity price λ using Algorithm 1, the optimal solution (i.e., the UE) is x^{L*} , let $\Delta^{TN} = \|x^{L*} - x^L\|$, $x^L \leftarrow x^{L*}$.
- 3: Update charging demand p^{dc} in (42) according to the obtained UE, and then solve relaxed OPF problem (38), retrieve LMP vector λ^* at the OPF solution, let $\Delta^{PDN} = \|\lambda^* - \lambda\|$, $\lambda \leftarrow \lambda^*$.
- 4: If $\Delta^{TN} + \Delta^{PDN} \leq \varepsilon$, terminate, and return the UE, OPF solution, and LMP as the NTE; else if $k = K_{it}^M$, terminate and report that the algorithm fails to converge;
- 5: Update $k = k + 1$, go to step 2.

3) *Uniqueness*: Because electricity prices appeared in TAP (16) are dual variables of the OPF problem, provable uniqueness guarantee from existing theories is non-trivial. Motivated by the stability condition (44), we conjecture that monotonic and bounded price sensitivity might lead to a unique NTE, i.e.,

$$\left\{ \frac{\partial \lambda_a^j}{\partial p_j^{dc}} > 0, \frac{\partial \lambda_a^j}{\partial p_j^{dc}} \left| \frac{\partial x_a}{\partial \lambda_a^j} \right| < E_B^{-1}, \forall a \in T_A^C \right\} \quad (45)$$

$$\forall p^{dc} \in P^{DC}$$

Under condition (45), $\mathcal{S}_{TAP}(\mathcal{S}_{OPF}(\cdot))$ is likely to be contractive, so the fixed point will be unique. The uniqueness of OPF solution under fixed p^{dc} can be expected due to the geometric property of power flow in radial networks revealed in [75]. However, condition (45) would be restrictive in general, and the LMP may be discontinuous, thus the partial derivative may not even exist. In fact, even if (45) is not satisfied, the NTE may still be unique.

B. The Best Response Decomposition Algorithm

In view of the interaction illustrated in Fig. 6, the NTE can be computed from a best-response decomposition procedure shown in Algorithm 2. The advantage relies on the convexity of TAP (16) and relaxed OPF problem (38), both of which can be solved reliably and efficiently.

Algorithm 2 converges provided that the initial value is close enough to the NTE, if there is any; otherwise, it may fail to converge, even an equilibrium exists. Practical treatments which facilitate convergence are discussed.

- 1) *Initialization*. When the penetration level of EV is low, p^{dc} only accounts for a small portion of the PDN load, we can initialize the LMP vector λ by solving problem (38) with $p^{dc} = \mathbf{0}^{|E_B|}$. Otherwise, we assume that each FCS serves the same amount of EVs (no matter whether this pattern is feasible) and calculate p^{dc} accordingly, and then solve problem (38) and recover an initial value of LMP.
- 2) *System data*. Properly widening the bounds of bus voltages and line flow limits or relaxing certain constraints in Cons-BND without jeopardizing a secure operation

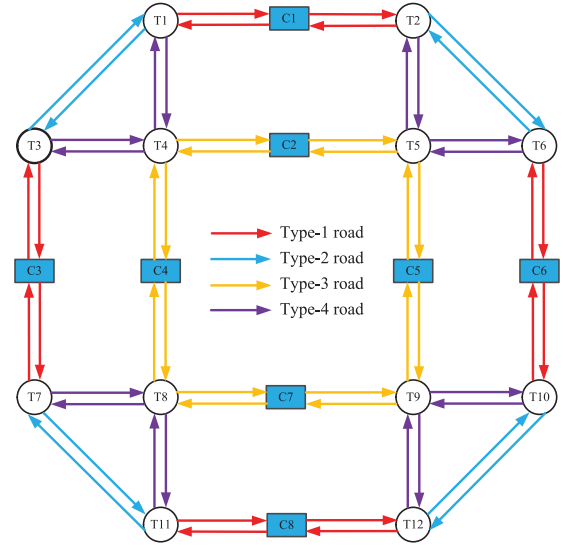


Fig. 7. Topology of the ring expressway system.

would generally contribute to more smooth LMPs and help Algorithm 2 converge. Moreover, stability criterion (45) implies that the NTE is more likely to be stable, if E_B is smaller, and the LMP curve $\lambda(p^{dc})$ is flatter. Since the marginal production cost at bus i is $2a_i p_i^g + b_i$, suppose the load increment is always supplied by the unit with the largest a_i , sensitivity $\partial \lambda_a^i / \partial p_i^{dc}$ is bounded by $2a_m$, where $a_m = \max_i \{a_i\}$, $\forall i$. In these regards, Algorithm 2 may converge faster with smaller a_m and E_B , and the NTE will be more stable in such circumstance.

- 3) Restricting the change of variables in two successive steps may accelerate convergence if the NTE exists but is not stable enough, i.e.,

$$\left| \frac{\partial \lambda_a^i}{\partial p_{di}^c} \left| \frac{\partial x_a}{\partial \lambda_a^i} \right| E_B \right| \approx 1$$

for some links.

V. CASE STUDIES

A. Basic Settings

This section exhibits experimental results on a test system with an emblematic ring expressway TN and a radial PDN. The topology of the TN is shown in Fig. 7, which consists of four classes of roads and eight FCSs in the outer and inner loops. Road data is given in Table III. The base value of traffic flow is 100 vehicles/hour. In the node-link module of FCS, the fixed charging time of link $a \in T_A^C$ is $t_a^0 = 20\text{min}$, the capacity is $c_a = 15$; For the bypass link, $t_a^0 = 0$, and c_a is equal to the capacity of the adjacent regular link, i.e., bypass link capacities of C1, C3, C6, C8 are 100, and those of C2, C4, C5, C7 are 80. O-D pairs and their traffic demands are listed in Table IV, indicating that the main traffic goes from the northwest to the east and the south. The monetary value of travel time is $\omega = 10\$/\text{h}$.

The PDN topology is shown in Fig. 8. The base value of power is 100MVA. Distribution line data is provided beside

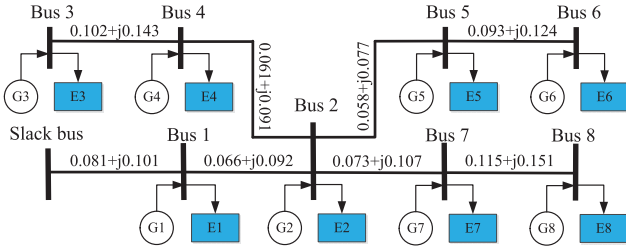


Fig. 8. Topology of the PDN.

TABLE III
PARAMETER OF ROADS

Road	Type-1	Type-2	Type-3	Type-4	Charging
c_a (p.u.)	100	100	80	60	15
t_a^0 (min)	5	8	5	7	20

TABLE IV
O-D PAIRS AND THEIR TRIP RATES (IN P.U.)

O-D pair	$q_g^{r,s}$	$q_e^{r,s}$	O-D pair	$q_g^{r,s}$	$q_e^{r,s}$
T1-T6	30	10	T3-T6	30	10
T1-T10	60	5	T3-T10	50	5
T1-T11	40	10	T3-T11	40	10
T1-T12	40	5	T3-T12	50	5
T4-T9	50	10	T4-T10	40	10
T4-T12	40	10			

each branch. Apparent power flow capacity is $S_l = 1.5$ for all lines. The fixed part of electrical demand at each bus is $p_j^{dt} = 0.1$ and $q_j^d = 0.1$. Voltage data are given by $\sqrt{U_j^n} = 0.88$, $\sqrt{U_j^m} = 1.05$, and $\sqrt{U_0} = 1.04$. Generators G1-G8 share the same data: $p_j^{gn} = 0$, $p_j^{gm} = 1.5$, $q_j^{gn} = -1.0$, $q_j^{gm} = 1.0$, $a_i = 0.3\$/\text{MW}^2\text{h}$, $b_i = 150\$/\text{MWh}$. The contract price with the main grid is $\rho = 140\$/\text{MWh}$. The energy demand of each EV is $E_B = 50\text{kWh}$. The active power demand of FCS is given by (28).

B. Results

Algorithm 2 is applied to compute the NTE. TAP (16) is solved by Baron [64], and OPF problem (38) in SOCP form is solved by Mosek [70]. The convergence tolerance is set as $\varepsilon = 0.5$. The maximal number of iterations is $K_{it}^M = 30$. The initial LMP is given by the OPF result with zero charging demands. Under above settings, Algorithm 2 terminates in 4 iterations. Results are shown in Figs. 9–10 and Tables V–VII. If Algorithm 2 is not interrupted when the convergence tolerant is met, the gap $\Delta^{TN} + \Delta^{PDN}$ generated in the first ten iterations is shown in Fig. 9. It is equal to 700.26 in the first iteration, and monotonically decreases with a value less than ε after the fourth iteration.

The UE pattern in the TN is depicted in Fig. 10, which illustrates that most vehicles travel on ring expressways in the inner and outer loops, because they have larger capacities and fewer traffic signals (indicated by smaller t_a^0). Trip information for each O-D pair is listed in Table V. Similar to the situation in Section II-E, EVs spend more time and cost than GVs due

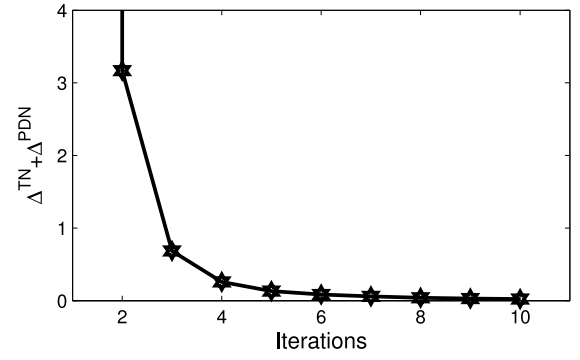


Fig. 9. Convergence performance of Algorithm 2.

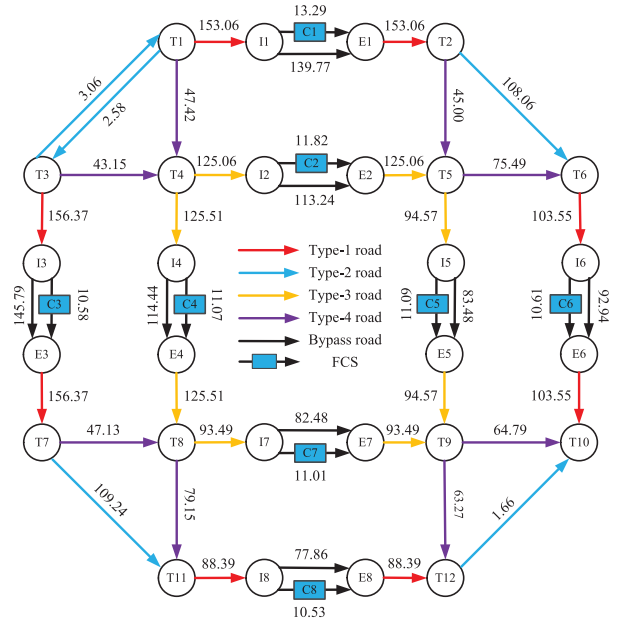


Fig. 10. Traffic flows at the UE pattern.

TABLE V
TRIP INFORMATION

O-D pair	Travel time (min)		Travel expense (\$)	
	GV	EV	GV	EV
1-6	27.87	55.66	4.6448	17.6728
1-10	39.59	62.01	6.5989	19.6268
1-11	36.68	59.07	6.1129	19.1409
1-12	46.79	69.63	7.7986	20.8265
3-6	35.87	59.58	5.9781	19.0060
3-10	47.59	69.95	7.9322	20.9601
3-11	28.68	51.07	4.7796	17.8076
3-12	39.59	61.95	6.5989	19.6268
4-9	31.89	54.64	5.3142	18.3421
4-10	40.31	62.73	6.7188	19.7467
4-12	40.18	62.54	6.6973	19.7252

to the waiting time and payment for recharging. If the refueling cost is taken into account, the average travel expense of GVs would be typically higher than that of EVs due to the higher price of petrol. The LMPs, active power demands at FCSs, and optimal generation dispatch strategies are exhibited

TABLE VI
LMPs, CHARGING DEMANDS, AND GENERATOR DISPATCH

Node	LMP(\$/MWh)	p_j^{dc} (MW)	p_j^g (MW)	q_j^g (MVar)
1	167.92	66.47	29.87	12.63
2	181.52	59.08	52.53	10.39
3	185.91	52.91	59.84	9.98
4	184.49	55.37	57.48	10.03
5	184.44	55.44	57.41	10.03
6	185.84	53.04	59.73	09.98
7	184.70	55.04	57.83	10.03
8	186.04	52.65	60.07	9.98

TABLE VII
OPF AT THE NTE

Node	V_j (p.u.)	Line	P_{ij}^l (MW)	Q_{ij}^l (MVar)
1	0.9520	0-1	105.89	10.57
2	0.9154	1-2	50.81	2.63
3	0.9045	2-4	11.05	0.136
4	0.9080	4-3	3.08	0.033
5	0.9081	2-5	11.45	0.124
6	0.9047	5-6	3.33	0.035
7	0.9075	2-7	9.88	0.127
8	0.9042	7-8	2.59	0.029

TABLE VIII
LMP AND CONVERGENCE RATE UNDER DIFFERENT E_B

E_B (kWh)		20	30	40	60
LMP	1	155.64	159.57	163.67	172.32
	2	163.98	169.72	175.56	187.59
	3	167.22	173.48	179.71	192.12
	4	165.98	172.09	178.27	190.79
	5	165.93	172.05	178.22	190.77
	6	167.13	173.39	179.63	192.12
	7	166.17	172.31	178.48	190.92
	8	167.42	173.68	179.87	192.05
Iterations		3	3	3	6

in Table VI. Two facts can be observed: 1) Since reverse power flow is banned, the bus voltage decreases and LMP increases from the root bus to the leaf bus; 2) EV flows in C3, C6, and C8 are smaller than those in the remaining FCSs, because they connect to leaf buses, where electricity are more expensive due to higher losses to deliver unit power from the slack bus. Meanwhile, the rationality of minimum travel expense constitutes a spontaneous demand response, preventing LMPs in C3, C6, C8 from rising higher. OPF results in PDN are provided in Table VII. From Tables VI–VII we can see that all security constraints are maintained. The energy consumption in C1 is mainly delivered from the slack bus, and LMP in C1 is the lowest due to the cheaper electricity price at the slack bus.

Finally, we investigate the impact of EV energy demand E_B on electricity prices. LMPs at NTEs associated with different values of E_B are enumerated in Table VIII. Clearly, LMPs rise when E_B grows larger, meanwhile, the gap between the highest and the lowest LMPs is also widened. The number of iterations in each scenario is also given in Table VIII

(4 for $E_B = 50$ kWh, which can be observed from Fig. 9). In general, when E_B grows, more iterations are needed before Algorithm 2 converges, because the interdependency between the two networks appears to be stronger.

VI. CONCLUSION

Electrification of the transportation sector creates interdependency across the transportation and power distribution networks. This paper presents a holistic modeling framework to study the intertwined traffic and power flows in the coupled networks. A multi-community user equilibrium model is proposed by jointly considering rational route choices, fast charging stations, and marginal pricing of electricity. It is revealed that the coupled network will reach an equilibrium state, which is characterized by a fixed-point problem. Built upon the convex optimization models of TAP and OPF, a best-response decomposition algorithm is suggested to identify the network equilibrium. The proposed methodology may serves as the foundation for analyzing, designing, and operating the coupled transportation and power distribution infrastructures in smart cities. The transportation system model can be extended to consider different charging requests due to heterogeneous battery status, and various types of charging services by employing more virtual links in the charging station module. Elastic traffic demands and destination choices can be incorporated using traditional methods in [8] as well.

REFERENCES

- [1] J. A. P. Lopes, F. J. Soares, and P. M. R. Almeida, "Integration of electric vehicles in the electric power system," *Proc. IEEE*, vol. 99, no. 1, pp. 168–183, Jan. 2011.
- [2] M. D. Galus *et al.*, "Integrating power systems, transport systems and vehicle technology for electric mobility impact assessment and efficient control," *IEEE Trans. Smart Grid*, vol. 3, no. 2, pp. 934–949, Jun. 2012.
- [3] J. Xiong, K. Zhang, Y. Guo, and W. Su, "Investigate the impacts of PEV charging facilities on integrated electric distribution system and electrified transportation system," *IEEE Trans. Transport. Electrific.*, vol. 1, no. 2, pp. 178–187, Aug. 2015.
- [4] M. Kezunovic and S. T. Waller, "PHEVs and BEVs in coupled power and transportation systems," in *Transportation Technologies for Sustainability*, M. Ehsani, F.-Y. Wang, and G. L. Brosch, Eds. New York, NY, USA: Springer, 2016, pp. 851–869.
- [5] J. G. Wardrop, "Some theoretical aspects of road traffic research," *Proc. Inst. Civil Eng.*, vol. 1, no. 3, pp. 325–362, May 1952.
- [6] M. Beckmann, C. McGuire, and C. Winsten, *Studies in the Economics of Transportation*. New Haven, CT, USA: Yale Univ. Press, 1956.
- [7] H. Z. Aashtiani and T. L. Magnanti, "Equilibria on a congested transportation network," *SIAM J. Algebra Discr.*, vol. 2, no. 3, pp. 213–226, 1981.
- [8] Y. Sheffi, *Urban Transportation Network: Equilibrium Analysis With Mathematical Programming Methods*. Upper Saddle River, NJ, USA: Prentice-Hall, 1985.
- [9] N. Jiang, C. Xie, and S. T. Waller, "Path-constrained traffic assignment: Model and algorithm," *Transp. Res. Rec. J. Transp. Res. Board*, pp. 25–33, Apr. 2012.
- [10] T.-G. Wang, C. Xie, J. Xie, and T. Waller, "Path-constrained traffic assignment: A trip chain analysis under range anxiety," *Transp. Res. C Emerg. Technol.*, vol. 68, pp. 447–461, Jul. 2016.
- [11] N. Jiang and C. Xie, "Computing and analyzing mixed equilibrium network flows with gasoline and electric vehicles," *Comput. Aided Civil Infrastruct. Eng.*, vol. 29, no. 8, pp. 626–641, Sep. 2014.
- [12] N. Jiang, C. Xie, J. C. Duthie, and S. T. Waller, "A network equilibrium analysis on destination, route and parking choices with mixed gasoline and electric vehicular flows," *Euro J. Transp. Logistics*, vol. 3, no. 1, pp. 55–92, Jun. 2014.

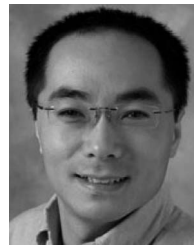
- [13] F. He, Y. Yin, and S. Lawphongpanich, "Network equilibrium models with battery electric vehicles," *Transp. Res. B Methodol.*, vol. 67, pp. 306–319, Sep. 2014.
- [14] M. R. Sarker, M. A. Ortega-Vazquez, and D. S. Kirschen, "Optimal coordination and scheduling of demand response via monetary incentives," *IEEE Trans. Smart Grid*, vol. 6, no. 3, pp. 1341–1352, May 2015.
- [15] I. Momber, A. Siddiqui, T. S. Román, and L. Söder, "Risk averse scheduling by a PEV aggregator under uncertainty," *IEEE Trans. Power Syst.*, vol. 30, no. 2, pp. 882–891, Mar. 2015.
- [16] W. Wei, F. Liu, and S. Mei, "Energy pricing and dispatch for smart grid retailers under demand response and market price uncertainty," *IEEE Trans. Smart Grid*, vol. 6, no. 3, pp. 1364–1374, May 2015.
- [17] Z. Yang *et al.*, "Profit maximization for plug-in electric taxi with uncertain future electricity prices," *IEEE Trans. Power Syst.*, vol. 29, no. 6, pp. 3058–3068, Nov. 2014.
- [18] S. I. Vagropoulos and A. G. Bakirtzis, "Optimal bidding strategy for electric vehicle aggregators in electricity markets," *IEEE Trans. Power Syst.*, vol. 28, no. 4, pp. 4031–4041, Nov. 2013.
- [19] M. G. Vayá and G. Andersson, "Optimal bidding strategy of a plug-in electric vehicle aggregator in day-ahead electricity markets under uncertainty," *IEEE Trans. Power Syst.*, vol. 30, no. 5, pp. 2375–2385, Sep. 2015.
- [20] W. Shuai, P. Maillé, and A. Pelov, "Charging electric vehicles in the smart city: A survey of economy-driven approaches," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 8, pp. 2089–2106, Aug. 2016.
- [21] B. Zhang and M. Kezunovic, "Impact on power system flexibility by electric vehicle participation in ramp market," *IEEE Trans. Smart Grid*, vol. 7, no. 3, pp. 1285–1294, May 2016.
- [22] H. Liu, Z. Hu, Y. Song, J. Wang, and X. Xie, "Vehicle-to-grid control for supplementary frequency regulation considering charging demands," *IEEE Trans. Power Syst.*, vol. 30, no. 6, pp. 3110–3119, Nov. 2015.
- [23] J. Donadee and M. D. Ilić, "Stochastic optimization of grid to vehicle frequency regulation capacity bids," *IEEE Trans. Smart Grid*, vol. 5, no. 2, pp. 1061–1069, Mar. 2014.
- [24] A. Y. S. Lam, K.-C. Leung, and V. O. K. Li, "Capacity estimation for vehicle-to-grid frequency regulation services with smart charging mechanism," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 156–166, Jan. 2016.
- [25] M. G. Vayá and G. Andersson, "Self scheduling of plug-in electric vehicle aggregator to provide balancing services for wind power," *IEEE Trans. Sustain. Energy*, vol. 7, no. 2, pp. 886–899, Apr. 2016.
- [26] A. Mohamed, V. Salehi, T. Ma, and O. Mohammed, "Real-time energy management algorithm for plug-in hybrid electric vehicle charging parks involving sustainable energy," *IEEE Trans. Sustain. Energy*, vol. 5, no. 2, pp. 577–586, Apr. 2014.
- [27] M. Yazdani-Damavandi *et al.*, "Modeling operational behavior of plug-in electric vehicles' parking lot in multienergy systems," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 124–135, Jan. 2016.
- [28] M. E. Khodayar, L. Wu, and M. Shahidehpour, "Hourly coordination of electric vehicle operation and volatile wind power generation in SCUC," *IEEE Trans. Smart Grid*, vol. 3, no. 3, pp. 1271–1279, Sep. 2012.
- [29] C. Liu, J. Wang, A. Botterud, Y. Zhou, and A. Vyas, "Assessment of impacts of PHEV charging patterns on wind-thermal scheduling by stochastic unit commitment," *IEEE Trans. Smart Grid*, vol. 3, no. 2, pp. 675–683, Jun. 2012.
- [30] J. Zhao, F. Wen, Z. Y. Dong, Y. Xue, and K. P. Wong, "Optimal dispatch of electric vehicles and wind power using enhanced particle swarm optimization," *IEEE Trans. Ind. Inform.*, vol. 8, no. 4, pp. 889–899, Nov. 2012.
- [31] A. D. Hilshey, P. D. H. Hines, P. Rezaei, and J. R. Dowds, "Estimating the impact of electric vehicle smart charging on distribution transformer aging," *IEEE Trans. Smart Grid*, vol. 4, no. 2, pp. 905–913, Jun. 2013.
- [32] M. E. Khodayar, L. Wu, and Z. Li, "Electric vehicle mobility in transmission-constrained hourly power generation scheduling," *IEEE Trans. Smart Grid*, vol. 4, no. 2, pp. 779–788, Jun. 2013.
- [33] M. Yilmaz and P. T. Krein, "Review of the impact of vehicle-to-grid technologies on distribution systems and utility interfaces," *IEEE Trans. Power Electron.*, vol. 28, no. 12, pp. 5673–5689, Dec. 2013.
- [34] R.-C. Leou, C.-L. Su, and C.-N. Lu, "Stochastic analyses of electric vehicle charging impacts on distribution network," *IEEE Trans. Power Syst.*, vol. 29, no. 3, pp. 1055–1063, May 2014.
- [35] E. Veldman and R. A. Verzijlbergh, "Distribution grid impacts of smart electric vehicle charging from different perspectives," *IEEE Trans. Smart Grid*, vol. 6, no. 1, pp. 333–342, Jan. 2015.
- [36] Q. Gong, Y. Li, and Z.-R. Peng, "Trip-based optimal power management of plug-in hybrid electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 57, no. 6, pp. 3393–3401, Nov. 2008.
- [37] R. Langari and J.-S. Won, "Intelligent energy management agent for a parallel hybrid vehicle-part I: System architecture and design of the driving situation identification process," *IEEE Trans. Veh. Technol.*, vol. 54, no. 3, pp. 925–934, May 2005.
- [38] Q. Wu *et al.*, "Driving pattern analysis for electric vehicle (EV) grid integration study," in *Proc. Innov. Smart Grid Technol. Conf. Europe*, Gothenburg, Sweden, Oct. 2010, pp. 1–6.
- [39] S. Bae and A. Kwasinski, "Spatial and temporal model of electric vehicle charging demand," *IEEE Trans. Smart Grid*, vol. 3, no. 1, pp. 394–403, Mar. 2012.
- [40] D. Tang and P. Wang, "Probabilistic modeling of nodal charging demand based on spatial-temporal dynamics of moving electric vehicles," *IEEE Trans. Smart Grid*, vol. 7, no. 2, pp. 627–636, Mar. 2016.
- [41] G. Wang, Z. Xu, F. Wen, and K. P. Wong, "Traffic-constrained multiobjective planning of electric-vehicle charging stations," *IEEE Trans. Power Del.*, vol. 28, no. 4, pp. 2363–2372, Oct. 2013.
- [42] W. F. Yao *et al.*, "A multi-objective collaborative planning strategy for integrated power distribution and electric vehicle charging systems," *IEEE Trans. Power Syst.*, vol. 29, no. 4, pp. 1811–1821, Jul. 2014.
- [43] S. Wang, Z. Dong, F. Luo, K. Meng, and Y. Zhang, "Stochastic collaborative planning of electric vehicle charging stations and power distribution system," *IEEE Trans. Ind. Inform.*, to be published.
- [44] S. D. Manshadi, M. E. Khodayar, K. Abdelghany, and H. Üster, "Wireless charging of electric vehicles in electricity and transportation networks," *IEEE Trans. Smart Grid*, to be published.
- [45] X. Dong, Y. Mu, H. Jia, J. Wu, and X. Yu, "Planning of fast EV charging stations on a round freeway," *IEEE Trans. Sustain. Energy*, vol. 7, no. 4, pp. 1452–1461, Oct. 2016.
- [46] H. Zhang, S. Moura, Z. Hu, and Y. Song, "PEV fast-charging station siting and sizing on coupled transportation and power networks," *IEEE Trans. Smart Grid*, to be published.
- [47] Y. Xiang *et al.*, "Economic planning of electric vehicle charging stations considering traffic constraints and load profile templates," *Appl. Energy*, vol. 178, pp. 647–659, Sep. 2016.
- [48] F. He, D. Wu, Y. Yin, and Y. Guan, "Optimal deployment of public charging stations for plug-in hybrid electric vehicles," *Transp. Res. B Methodol.*, vol. 47, pp. 87–101, Jan. 2013.
- [49] F. He, Y. Yin, J. Wang, and Y. Yang, "Sustainability SI: Optimal prices of electricity at public charging stations for plug-in electric vehicles," *Netw. Spatial Econ.*, vol. 16, no. 1, pp. 131–154, Mar. 2016.
- [50] W. Wei, S. Mei, L. Wu, M. Shahidehpour, and Y. Fang, "Optimal traffic-power flow in urban electrified transportation networks," *IEEE Trans. Smart Grid*, vol. 8, no. 1, pp. 84–95, Jan. 2017.
- [51] M. Alizadeh *et al.*, "Optimal pricing to manage electric vehicles in coupled power and transportation networks," *IEEE Trans. Control Netw. Syst.*, to be published.
- [52] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Trans. Power Del.*, vol. 4, no. 2, pp. 1401–1407, Apr. 1989.
- [53] M. E. Baran and F. F. Wu, "Optimal capacitor placement on radial distribution systems," *IEEE Trans. Power Del.*, vol. 4, no. 1, pp. 725–734, Jan. 1989.
- [54] M. E. Baran and F. F. Wu, "Optimal sizing of capacitors placed on a radial distribution system," *IEEE Trans. Power Del.*, vol. 4, no. 1, pp. 735–743, Jan. 1989.
- [55] M. Farivar and S. H. Low, "Branch flow model: Relaxations and convexification," *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 2554–2572, Aug. 2013.
- [56] M. Schneider, A. Stenger, and D. Goeke, "The electric vehicle-routing problem with time windows and recharging stations," *Transp. Sci.*, vol. 48, no. 4, pp. 500–520, Mar. 2014.
- [57] H. Yang *et al.*, "Electric vehicle route optimization considering time-of-use electricity price by learnable partheno-genetic algorithm," *IEEE Trans. Smart Grid*, vol. 6, no. 2, pp. 657–666, Mar. 2015.
- [58] Q. Guo, S. Xin, H. Sun, Z. Li, and B. Zhang, "Rapid-charging navigation of electric vehicles based on real-time power systems and traffic data," *IEEE Trans. Smart Grid*, vol. 5, no. 4, pp. 1969–1979, Jul. 2014.
- [59] E. A. Grunditz and T. Thiringer, "Performance analysis of current BEVs based on a comprehensive review of specifications," *IEEE Trans. Transport. Electrific.*, vol. 2, no. 3, pp. 270–289, Sep. 2016.
- [60] K. B. Davidson, "A flow-travel time relationship for use in transportation planning," in *Proc. Aust. Road Res. Board*, Melbourne, VIC, Australia, 1966, pp. 183–194.
- [61] W. Wei, S. Mei, L. Wu, J. Wang, and Y. Fang, "Robust operation of distribution networks coupled with urban transportation infrastructures," *IEEE Trans. Power Syst.*, vol. 32, no. 3, pp. 2118–2130, May 2017.

- [62] Bureau of Public Roads, "Traffic assignment manual," U.S. Dept. Commerce, Washington, DC, USA, 1964.
- [63] D. Ralph, "Mathematical programs with complementarity constraints in traffic and telecommunications networks," *Philos Trans. Roy. Soc. London A Math. Phys. Eng. Sci.*, vol. 366, no. 1872, pp. 1973–1987, 2008.
- [64] M. Tawarmalani and N. V. Sahinidis, "A polyhedral branch-and-cut approach to global optimization," *Math. Program.*, vol. 103, no. 2, pp. 225–249, Jun. 2005.
- [65] B. Subhonmesh, S. H. Low, and K. M. Chandy, "Equivalence of branch flow and bus injection models," in *Proc. 50th Annu. Allerton Conf. Commun. Control Comput. (Allerton)*, Monticello, IL, USA, Oct. 2012, pp. 1893–1899.
- [66] N. Li, L. Chen, and S. H. Low, "Exact convex relaxation of OPF for radial networks using branch flow model," in *Proc. IEEE 3rd Int. Conf. Smart Grid Commun.*, Tainan, Taiwan, Nov. 2012, pp. 7–12.
- [67] L. Gan, N. Li, U. Topcu, and S. H. Low, "Exact convex relaxation of optimal power flow in radial networks," *IEEE Trans. Autom. Control*, vol. 60, no. 1, pp. 72–87, Jan. 2015.
- [68] S. Huang, Q. Wu, J. Wang, and H. Zhao, "A sufficient condition on convex relaxation of AC optimal power flow in distribution networks," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 1359–1368, Mar. 2017.
- [69] W. Wei, J. Wang, N. Li, and S. Mei, "Optimal power flow of radial networks and its variations: A sequential convex optimization approach," *IEEE Trans. Smart Grid*, to be published.
- [70] (2016). *Mosek Website*. [Online]. Available: <https://www.mosek.com/>
- [71] A. Ben-Tal and A. Nemirovski. (2016). *Lectures on Modern Convex Optimization*. [Online]. Available: http://www2.isye.gatech.edu/~nemirovs/Lect_ModConvOpt.pdf
- [72] F. Li, "Continuous locational marginal pricing (CLMP)," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1638–1646, Nov. 2007.
- [73] J. Bonnans and A. Shapiro, *Perturbation Analysis of Optimization Problems*. New York, NY, USA: Springer, 2013.
- [74] K. C. Border, *Fixed Point Theorems With Applications to Economics and Game Theory*. Cambridge, U.K.: Cambridge Univ. Press, 1989.
- [75] J. Lavaei, D. Tse, and B. Zhang, "Geometry of power flows and optimization in distribution networks," *IEEE Trans. Power Syst.*, vol. 29, no. 2, pp. 572–583, Mar. 2014.



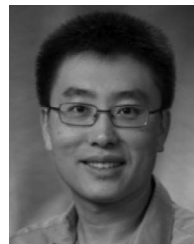
Wei Wei (M'15) received the B.Sc. and Ph.D. degrees in electrical engineering from Tsinghua University, Beijing, China, in 2008 and 2013, respectively.

He was a Post-Doctoral Researcher with Tsinghua University from 2013 to 2015. He was a Visiting Scholar with Cornell University, Ithaca, NY, USA, in 2014, and with Harvard University, Cambridge, MA, USA, in 2015. He is currently an Assistant Professor with Tsinghua University. His research interests include applied optimization and energy economics.



Lei Wu (SM'13) received the B.S. degree in electrical engineering and the M.S. degree in systems engineering from Xi'an Jiaotong University, Xi'an, China, in 2001 and 2004, respectively, and the Ph.D. degree in electrical engineering from the Illinois Institute of Technology (IIT), Chicago, IL, USA, in 2008.

From 2008 to 2010, he was a Senior Research Associate with the Robert W. Galvin Center for Electricity Innovation, IIT. He was a summer Visiting Faculty with NYISO in 2012. He is currently an Associate Professor with the Electrical and Computer Engineering Department, Clarkson University, Potsdam, NY, USA. His research interests include power systems operation and planning, energy economics, and community resilience microgrid.



Jianhui Wang (M'07–SM'12) received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology, Chicago, IL, USA, in 2007.

He is currently an Associate Professor with the Department of Electrical Engineering, Southern Methodist University, Dallas, TX, USA. He also holds a joint appointment as a Section Lead for Advanced Power Grid Modeling, Energy Systems Division, Argonne National Laboratory, Argonne, IL, USA.

Dr. Wang has held visiting positions in Europe, Australia, and Hong Kong including a VELUX Visiting Professorship with the Technical University of Denmark. He was a recipient of the IEEE Power and Energy Society (PES) Power System Operation Committee Prize Paper Award in 2015. He is the Secretary of the IEEE PES Power System Operations, Planning and Economics Committee. He is an Associate Editor of the *Journal of Energy Engineering* and an Editorial Board Member of *Applied Energy*. He is the Editor-in-Chief of the IEEE TRANSACTIONS ON SMART GRID and an IEEE PES Distinguished Lecturer.

Shengwei Mei (F'15) received the B.Sc. degree in mathematics from Xinjiang University, Urumqi, China, in 1984, the M.Sc. degree in operations research from Tsinghua University, Beijing, China, in 1989, and the Ph.D. degree in automatic control from the Chinese Academy of Sciences, Beijing, China, in 1996.

He is currently a Professor of Tsinghua University. His research interests include power system complexity and control, and game theory and its application in power systems.